Zewail City of Science, Technology and Innovation
University of Science and Technology
School of Computational Sciences and Artificial Intelligence

CSAI 203 - Fall 2026

Introduction to Software Engineering

<Online Quiz & Auto-Grading System>

<SRS>

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1. Introduction

1.1 Purpose

This document presents the Software Requirements Specification (SRS) for the Online Quiz & Auto-Grading System. It clearly defines the functional and non-functional requirements to serve as a shared reference for team members, instructors, and evaluation staff during the development, testing, and assessment phases of the project.

1.2 Scope

The Online Quiz & Auto-Grading System is a web-based educational platform designed to streamline and automate the process of academic evaluation. The system enables instructors to create question banks, build timed quizzes, publish assessments, and view auto-generated grade reports and analytics. For students, it provides a secure interface to attempt exams, track remaining time, review immediate feedback, and check past performance. The system aims to eliminate manual grading delays, reduce human error, and offer a responsive, user-friendly UI.

1.3 Definitions, Acronyms, and Abbreviations

- **SRS:** Software Requirements Specification.
- **UI:** User Interface.
- **MVC:** Model-View-Controller architecture pattern.
- **FR:** Functional Requirement.
- **NFR:** Non-Functional Requirement.
- **Question Bank:** A centralized repository where instructors store and categorize reusable exam questions.
- **Auto-Grading:** The process where the system automatically evaluates student submissions against correct answers upon test completion.

1.4 References

1. CSAI 203 Course Project Specifications, Fall 2026, Zewail City of Science and Technology.
2. IEEE Std 830-1998 (*IEEE Recommended Practice for Software Requirements Specifications*)

1.5 Overview

This document is divided into four primary sections:

- **Section 1 (Introduction):** Provides a high-level overview, document conventions, scope, and basic terms.
- **Section 2 (Overall Description):** Outlines the system perspective, user roles, operating conditions, constraints, and dependencies.
- **Section 3 (Specific Requirements):** Detailed specifications of functional and non-functional requirements, use cases, domain models, and interface constraints.
- **Section 4 (Appendices):** Contains supplementary details including the data dictionary and glossary.

2. Overall Description

2.1 Product Perspective

The Online Quiz & Auto-Grading System is a web-based application designed to facilitate the creation, management, and evaluation of online quizzes. It provides instructors with tools to create quizzes, manage question banks, add questions, set quiz durations, publish quizzes, and review student results. Students can use the system to access published quizzes, answer questions, submit their attempts, and view their scores and feedback.

The system is intended to operate as a standalone application within the scope of this project. Its main purpose is to simplify the online assessment process, reduce the effort required for manual grading, and provide students and instructors with convenient access to quiz information and results.

The system supports two primary user roles: instructors and students. Instructors are responsible for preparing and managing quizzes, while students use the system to take available quizzes and review their results. The system automatically grades submitted quiz attempts based on predefined correct answers.

Integration with external educational platforms or university management systems is not included in the current project scope.

2.2 Product Functions

Online Quiz & Auto-Grading System provides the following main functions:

1. User Registration and Login: Allows students and instructors to create accounts and access the system using their registered credentials.
2. User Logout: Allows logged-in users to securely end their active sessions.
3. Quiz Creation: Enables instructors to create quizzes by providing a title and adding questions.
4. Question Management: Allows instructors to add questions and their multiple-choice options to quizzes or question banks.
5. Question Bank Management: Enables instructors to create and manage question banks containing multiple-choice questions.
6. Quiz Timer : Allows instructors to set the duration of a quiz before publishing it.
7. Quiz Publishing: Enables instructors to publish quizzes that meet the required conditions, making them available to students.
8. Quiz Editing and Deletion: Allows instructors to edit or delete their own eligible quizzes, question banks, and questions according to the system's rules.
9. Quiz Taking: Allows students to access published quizzes and start quiz attempts.
10. Answer Selection and Modification: Enables students to select answers and change their selections while their quiz attempts remain active.
11. Timer Display: Displays the remaining time during a timed quiz attempt.
12. Quiz Submission: Allows students to submit their attempts manually, while the system automatically submits attempts when the time expires
13. Automatic Grading: Evaluates submitted answers by comparing them with the correct answers stored in the system.
14. Score Calculation and Display: Calculates the total score based on correct answers and displays the result to students.
15. Result and Feedback Display: Allows students to review their submitted answers, correct answers, scores, and any available instructor feedback.
16. Grade Reports: Enables instructors to review students scores and view a summary of the score distribution for their quizzes.
17. Profile Viewing: Allows users to view their basic account information and relevant information associated with their roles.
18. Assessment Navigation: Allows students to move between questions in an assessment while retaining their previously selected answers.

2.3 User Classes and Characteristics

The system has two primary user classes: instructors and students.

2.3.1 Instructors:

Instructors are responsible for creating and managing quizzes and question banks. They can add multiple-choice questions, configure quiz durations, publish quizzes, manage eligible content, and review student performance through grade reports.
Instructors are expected to have basic computer skills and familiarity with using web applications. They should understand how to prepare questions, identify correct answers, and evaluate student performance.

2.3.2 Students:

Students use the system to access published quizzes, start quiz attempts, select answers, navigate between questions, monitor the remaining time, and submit their attempts. They can also view their scores, correct answers, and available feedback after submission.
Students are expected to have basic computer skills and the ability to navigate a web-based application.

Both user classes require valid accounts and appropriate access permissions to use the system's features.

2.4 Software Stakeholders

The main stakeholders of the Online Quiz & Auto-Grading System are:

2.4.1 Students:

Students are direct users of the system. Their primary interests are accessing available quizzes, answering questions easily, receiving accurate results, and reviewing feedback on their performance.

2.4.2 Instructors:

Instructors use the system to prepare assessments, manage questions, publish quizzes, and monitor student performance. They require reliable quiz management features and accurate automatic grading.

2.4.3 Educational Institution:

The educational institution may benefit from the system by supporting online assessment activities, reducing manual grading efforts, and providing a more organized way to manage quiz results.

2.4.4 Project Team:

The project team is responsible for designing, developing, testing, and maintaining the system as part of the CSAI 203 Introduction to Software Engineering course.
The team must ensure that the system meets the agreed requirements.

2.5 Operating Environment

The Online Quiz & Auto-Grading System is intended to operate as a web-based application.

The expected operating environment includes:

- Hardware: Desktop computers, laptops, tablets, or smartphones capable of accessing a web application.
- Operating Systems: Common operating systems that support a compatible web browser, including Windows, Linux, Android, and iOS.
- Web Browsers: Modern web browsers that support the technologies used to implement the application.
- Network: An internet connection is required to access the application's online features.
- Software Environment: The application will run using the programming language, web framework, database, and hosting environment selected by the project team.

The exact technical requirements will depend on the implementation technologies chosen for the project.

2.6 Design and Implementation Constraints

1. Course Requirements:
   The system must comply with the requirements and submission guidelines of the CSAI 203 Introduction to Software Engineering course.
2. Project Scope:

The implementation must focus on the agreed features of the Online Quiz & Auto-Grading System.

3. User Roles and Permissions: The system must distinguish between instructors and students and restrict access to functions according to their roles.
4. Automatic Grading: The system must calculate quiz scores according to the predefined correct answers and scoring rules.
5. Quiz Timing: Quiz durations must be positive, and the system must enforce the configured time limit for active quiz attempts.
6. Data Protection: The system must protect user account information, quiz content, submitted answers, and student results from unauthorized access.
7. Content Management Rules: Instructors must only manage their own content, and restrictions must be applied to questions belonging to published quizzes.
8. Implementation Technologies: The programming language, framework, database, and other technical components must be selected by the project team and must support the required system functions.
9. Academic Deadlines: Development and documentation must be completed according to the course deadlines and submission requirements.

2.7 User Documentation

The system is expected to provide basic user documentation to help instructors and students understand how to use its main features.

The documentation may include a user guide and a Frequently Asked Questions (FAQ) section covering the following topics:

- Creating an account and logging in.
- Navigating the system and viewing user profiles.
- Creating and managing question banks.
- Creating quizzes and adding questions.
- Setting quiz durations and publishing quizzes.
- Accessing and completing quizzes.
- Selecting answers and navigating between questions.
- Submitting quiz attempts and understanding automatic submission.
- Viewing quiz scores, correct answers, and available feedback.
- Accessing grade reports, where applicable.

The documentation should be written clearly and organized so that users with basic computer skills can understand the system's operation.

2.8 Assumptions and Dependencies

The development and operation of the Online Quiz & Auto-Grading System depend on the following assumptions:

1. Internet Availability: Users have a working internet connection when accessing the web-based application.
2. Compatible Devices and Browsers: Users access the system through supported devices and compatible web browsers.
3. Valid User Information: Users provide the required information when registering and use valid credentials when logging in.
4. Instructor-Provided Content: Instructors provide valid quiz questions, multiple-choice options, correct answers, and any required feedback.
5. Question Bank and Quiz Management: Instructors create and manage their own question banks and quizzes according to the system's rules.
6. Automatic Grading Rules: The system relies on stored correct answers and predefined scoring rules to calculate quiz results.
7. Data Storage: The application requires an appropriate data storage mechanism to maintain user accounts, quizzes, questions, attempts, answers, and results.
8. Software Dependencies: The application depends on the selected programming language, framework, database, and hosting environment functioning correctly.
9. System Availability: Users can access the system when the application and its required services are operational.
10. Project Scope: Integration with external learning management systems or university platforms is not assumed.

These assumptions may be reviewed and updated if the project's requirements or implementation approach changes.

3. Specific Requirements

3.1 Functional Requirements

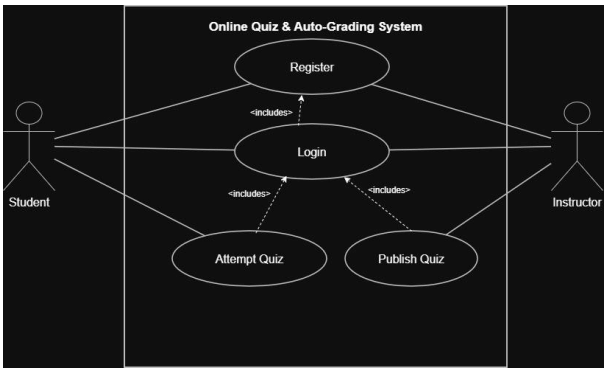
Function	Description	Function	Description
Function 1: Registration/ Login	The system allows a student or an instructor to create an account by entering a name, a unique email address, a password, and a selected role. Registered users will be able to log in with valid credentials.	Function 11: Edit	The system will allow instructors to edit their own question banks, and unpublished quizzes. Questions belonging to a published quiz will not be modified.
Function 2: Logout	The system will allow logged-in users to log out of their account.	Function 12: Delete	The system will allow instructors to delete their own question banks/quizzes or questions in them and unpublished quizzes.
Function 3: Add questions	This function allows instructors to add questions to either their quiz or question bank. Each question contains a question statement, four options, and one correct option; the same quiz/question bank should not contain the same question twice.	Function 13: Publish	The system will allow an instructor to publish a quiz only when it has at least one valid question and a positive duration. Publishing will make it visible to students.
Function 4: Quiz duration	The system will allow the instructor to enter a positive number of minutes as the quiz duration before publication. The chosen duration will be stored and applied when a student starts the quiz.	Function 14: Take quiz	The system will allow a logged-in student to open a published quiz and begin an attempt. Once begun, the questions will be shown and the attempt start time will be recorded.
Function 5: Submit	The system will allow a student to submit an active quiz attempt. On submission, selected answers will be stored, the attempt will be marked complete. If the timer expires, the system automatically submits the attempt.	Function 15: Take assessments	The system will allow a student to move through the questions in an assessment which is a question bank/quiz without losing previously selected answers.
Function 6: Display results	The system will display the quiz result after instant grading, including the quiz title, number of questions, correct answers, and total score.	Function 16: Select answers	The system will allow the student to select exactly one option for each multiple-choice question in quiz or assessment.
Function 7: View profile	The system will allow a logged-in user to view their name, email address, and user role from a profile page. For Instructors their profile will include published quizzes/question banks, while for students each student will see their attempted assessments and quizzes with grades for each.	Function 17: View score	The system will calculate the total score from one point per correct answer and zero points per incorrect or unanswered question. The resulting numeric score will be shown to the student and included in the instructor report.
Function 8: Grade reports	The system will allow an instructor to open a report for one of their quizzes, view each submitted student score, and see a simple grade distribution showing how scores are spread across attempts.	Function 18: View timer	The system will show the remaining time during an attempt in minutes and seconds. It shall calculate the countdown from the attempt start time and configured duration; reaching zero will trigger automatic submission.
Function 9: Create	The system will allow an instructor to create a named question bank and store their own multiple-choice questions inside it.	Function 19: Change answers	The system will allow a student to replace a selected option while the attempt is active and time remains which is then used to grade.

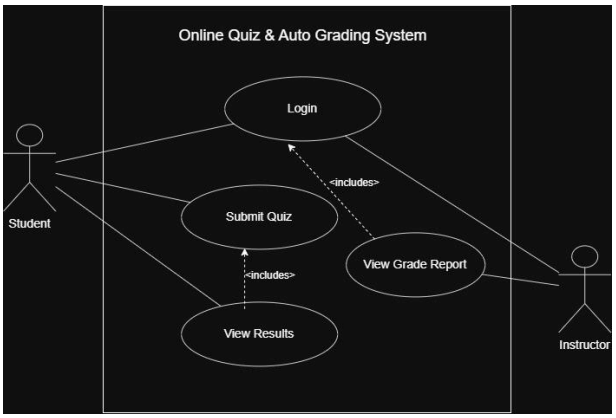
question banks			
Function 10: Create quiz	The system will allow a logged-in instructor to create a quiz by entering a title and questions. A newly created quiz will remain unpublished until the instructor chooses to publish it.	Function 20: View feedback	The system will allow students to view their selected answer and the correct answer for each question after submission and any added feedback from the instructor later.

Table 1. Functional Requirements

3.2 Use Case Model

3.2.1 Use Case Diagram





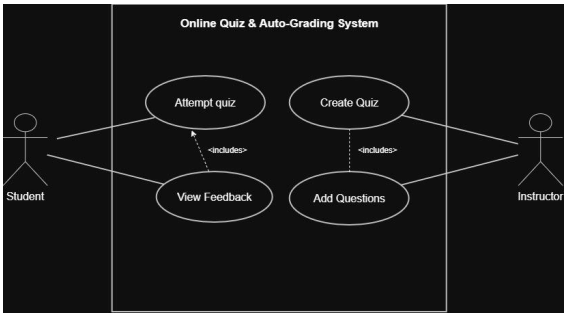


Fig 1,2,3. Use cases

3.2.2 Use Case Descriptions

Use Case	Main Flow	Alternative Flows	Preconditions	Postconditions	Actors	Description
1.Registration/Login	1. The user selects Registration or Login. 2. For registration, user enters name, unique email,	If the registration email already exists, registration is rejected. If login credentials are invalid, login is rejected.	Users can access the application.	An account is created or a	Student, Instructor	User registers/logs in

	password, and selected role; for login, user enters email and password. 3. The system validates the entered information. 4. The system creates the account.			registered user is logged in.		
2. Add questions	1. The instructor selects a quiz or question bank. 2. The instructor enters the question statement and four options. 3. The instructor identifies one correct option. 4. The system checks for a duplicate question in the selected quiz or bank. 5. The system adds the valid question.	If the question is incomplete or already present in the same quiz or bank, the system does not add it.	The instructor is logged in and has an editable quiz or question bank.	Questions are stored in the selected quiz or question bank.	Instructor	Instructor adds questions to quiz
3. View results	1. System obtains the graded result. 2. System displays quiz title, number of questions, correct answers, and total score.	If no graded result is available, the result cannot yet be displayed.	The submitted attempt has been graded.	The student can see the quiz result.	Student	Student views their results in quiz
4. Create quiz	1. Instructor selects Create quiz. 2. The instructor enters a title and questions. 3. The system creates and stores the new quiz as unpublished.	If required quiz information has not been provided, the quiz cannot be completed as described.	The instructor is logged in.	An unpublished quiz exists.	Instructor	Instructor creates quiz
5. Publish	1. The instructor selects an unpublished quiz. 2. Instructor selects Publish. 3. The system verifies at least one valid question and a positive duration. 4. The system publishes the quiz for students.	If the quiz has no valid question or lacks a positive duration, publication is blocked.	The instructor is logged in and owns an unpublished quiz.	The quiz is published and visible to students.	Instructor	Instructor publish quiz after creation
6. Attempt quiz	1. The student selects a published quiz. 2. A student begins an attempt. 3. The system records the attempt start time. 4. The system shows the questions.	If the quiz is not published, it is not available to start.	Students are logged in; the quiz is published.	An active attempt has started.	Student	Student attempts to take a quiz

7. View feedback	1. Students open feedback for the submitted attempt. 2. The system shows the student selection and correct option for each question. 3. The system shows any instructor feedback that has been added.	If no instructor feedback has been added, only selected and correct answers are displayed.	The student has submitted the relevant attempt.	The student can review available feedback.	Student	Student views feedback from instructor

Table 2. Class Descriptions

3.3 Domain Model

3.3.1 Conceptual Class Diagram

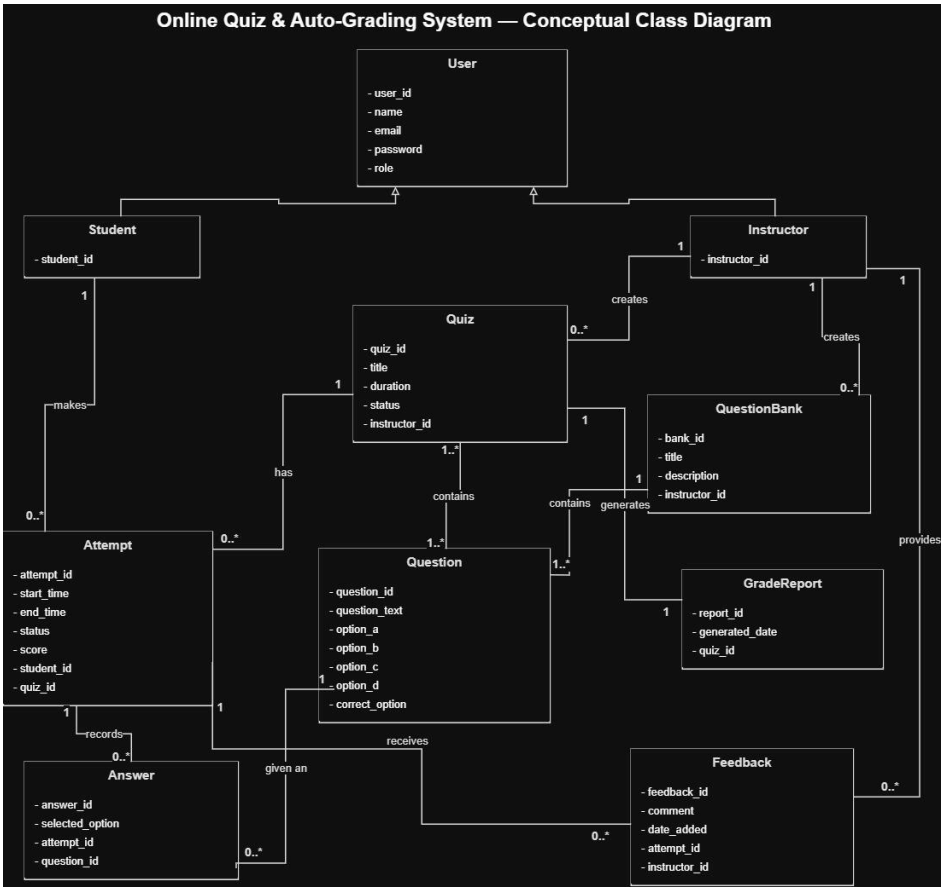


Fig 4. Class Diagram

3.3.2 Class Descriptions

Class Name	Description	Key Attributes	Associations
User	Stores the basic information of all users in the system.	user_id, name, email, password, role	Student and Instructor inherit from User.
Student	Represents students who take quizzes, submit answers, and view their results.	student_id	Inherits from User, Makes Attempts.
Instructor	Represents instructors who create quizzes, manage questions, and view student grades.	instructor_id	Inherits from User, Creates Quizzes, Creates Question Banks, Provides Feedback.
QuestionBank	Stores a collection of multiple-choice questions created by an instructor.	bank_id, title, description, instructor_id	Created by Instructor, Contains Questions.
Question	Stores the question text, four answer options, and the correct answer.	question_id, question_text, option_a, option_b, option_c, option_d, correct_option	Contained in Question Bank, and Quizzes, Has Answers.
Quiz	Stores quiz information, including its title, duration, and publication status.	quiz_id, title, duration, status, instructor_id	Created by Instructor, Contains Questions, Has Attempts, Generates Grade Report.
Attempt	Stores information about a student's quiz attempt, including the starting time and score.	attempt_id, start_time, end_time, status, score, student_id, quiz_id	Made by Student, Belongs to Quiz, Records Answers, Receives Feedback.
Answer	Stores the answer selected by a student for a question during a quiz attempt.	answer_id, selected_option, attempt_id, question_id	Recorded in Attempt, Given for Question.
Feedback	Stores quiz score and additional feedback from instructors to help students understand their answers.	feedback_id, comment, date_added, attempt_id, instructor_id	Provided by Instructor, Received by Attempt.
GradeReport	Shows student scores and grade distributions for a quiz.	report_id, generated_date, quiz_id	Generated by Quiz.

Table 3. Class description

3.4 Non-Functional Requirements

NFR-01: Fast Response Time
The system should respond to user actions, such as opening quizzes and viewing results, within two seconds under normal conditions.

Testing: Measure the time required to perform these actions.

NFR-02: Accuracy
The system should save answers correctly and calculate quiz scores accurately. Each correct answer receives one point, while incorrect or unanswered questions receive zero points.

Testing: Compare calculated scores with expected results.

NFR-03: Performance
The system should support multiple students using it at the same time.

Testing: Test the system with multiple users simultaneously.

NFR-04: Usability
The system should be easy for students and instructors to use.

Testing: Ask users to complete common tasks and evaluate how easily they can use the system.

NFR-05: Security
The system should protect user accounts and quiz results and prevent unauthorized access.

Testing: Check that students cannot access instructor-only features and vice versa.

3.5 External Interface Requirements

3.5.1 User Interface

The system shall provide a simple web interface with the following pages:

- Registration and Login: Allow users to create accounts and log in.
- Dashboard: Display available features according to the user's role.
- Quiz Management: Allow instructors to create, edit, and publish quizzes.
- Question Bank: Allow instructors to manage their questions in each question bank.
- Quiz Page: Allow students to answer questions and view the timer.
- Results Page: Display quiz scores and feedback.
- Profile Page: Display user information based on role if it's an instructor it will display their information (email, username) and their quizzes and questions banks, if it's a student it will display their information (email, username) and their taken quizzes and attempted assessments.

3.5.2 Hardware Interface

The system shall work on computers, laptops, tablets, and smartphones. Users will interact with the system using a screen, keyboard, mouse, or touchscreen. No special hardware is required

3.5.3 Software Interface.

The system shall use a web browser to display the application and a database to store user accounts, quizzes, questions, answers, and results. The application will use Python and Flask, with the database technology selected by the project team.

3.5.4 Communication Interface

The system shall use an internet connection to exchange data between users' browsers and the application server. HTTP/HTTPS will be used for communication, and HTTPS should be used when the system is deployed to protect user information.

4. Appendices

4.1 Appendix A: Data Dictionary

Table 1: Functional Requirements: Describes the main functions and features provided by the Online Quiz & Auto-Grading System.

Table 2: Use Case Descriptions: Describe each use case, including its basic flow, alternative flows, preconditions, and postconditions.

Table 3: Class Description: Describes the main classes of the system, including their purposes, attributes, and relationships.

Figure 1: Account Management Use Case Diagram: Illustrates how students and instructors interact with the system to manage their accounts.

Figure 2: Instructor Quiz Management Use Case Diagram: Illustrates how instructors interact with the system to create and manage quizzes, questions, and grade reports.

Figure 3: Student Attempts and Results Use Case Diagram: Illustrates how students interact with the system to take quizzes, submit answers, and view results and feedback.

Fig. : Class Diagram: Illustrates the main classes of the system and the relationships between them, including inheritance and associations.

4.2 Appendix B: Glossary

Term	Definition
SRS	Software Requirements Specification; a document describing the system's functional and non-functional requirements.
Functional Requirement (FR)	A function or feature that the system must provide.
Non-Functional Requirement (NFR)	A requirement describing system qualities such as performance, security, usability, and accuracy.
UI (User Interface)	The web pages through which students and instructors interact with the system.
MVC	Model-View-Controller; a software architecture pattern.
Multiple-Choice Question (MCQ)	A question containing four answer options, with one correct option.
Auto-Grading	Automatically evaluating submitted answers against stored correct answers.
HTTP/HTTPS	Communication protocols used to exchange data between web browsers and the application server.